HOSTEL MANAGEMENT SYSTEM

SYSTEM TESTING REPORT

Hostel Management System

Quaid-i-Azam University

Software Testing Documentation

Document status	Prepared test plan and test cases; execution results pending
Version / date	1.0 | 26 October 2026
System context	Hostel Management System (HMS), Quaid-i-Azam University
Report basis	Project workflows taken from supplied performance-testing report; implementation details not supplied

Hostel Management System - Quaid-i-Azam University

Document Control

Field	Details
Document title	System Testing Report
Project	Hostel Management System
Institution	Quaid-i-Azam University
Document version	1.0
Prepared for	Project testing and report documentation
Testing status	Planned test cases documented; empirical execution results not supplied
Repository destination	reports/system-testing-report.docx on the report branch

1. Introduction

This report defines system testing for the Hostel Management System (HMS) described in the supplied project report. System testing checks complete user-facing workflows across the application and its data storage, including student and staff activities. The actual system, requirements, and execution evidence were not provided; this document is a test plan and makes no claim that tests have been run.

2. Objectives

• Verify end-to-end workflows for student, warden, and administrator roles.

• Confirm that data remains accurate as records move through application, approval, allocation, complaint, fee, and visitor workflows.

• Check validation, status transitions, search/filter behavior, and user-visible errors.

• Confirm reports and dashboards agree with underlying test records.

• Record repeatable results, defects, evidence, and retest status.

3. Scope

System area	System-level coverage
Login and dashboards	Authenticate each role and verify correct landing page, navigation, and role-specific information.
Hostel applications	Submit, review, approve/reject, search, and inspect application status.
Rooms and bed allotment	Search availability, allot eligible students, and verify room/bed occupancy updates.
Complaints	Submit complaint, view status, update through authorized workflow, and confirm history.
Fees	Retrieve fee status and record/verify test payment status where supported.
Visitors	Record visitor entry/exit and search visitor history.
Notices	Create/publish where supported and verify notice visibility for intended audience.
Reports	Generate occupancy and other available reports and reconcile output to test data.
Cross-functional checks	Verify that changes appear consistently across dashboards, searches, details, and reports.

Out of scope: unit and component tests, load/performance certification, security penetration testing, production data migration, and third-party service certification. These require separate plans and evidence.

4. Test Environment and Preconditions

Environment item	Value to record before testing
Application build / commit	_____
Application URL / deployment type	_____
Browser / device / OS	_____
Database engine and version	_____
Test accounts and roles	_____
Test-data volume and reset method	_____
Test date and tester	_____

Preconditions

• Use an approved non-production environment and synthetic test data.

• Prepare student, warden, and administrator accounts and representative records for hostels, rooms, beds, applications, complaints, fees, visitors, and notices.

• Document expected outcomes and permitted status transitions with the project team before execution.

• Confirm backup/reset steps for tests that create or update records.

• Capture screenshots or record identifiers for evidence; remove or mask personal data in shared results.

5. System Test Cases

The following cases are proposed and marked Not run. Confirm feature availability and expected rules against the implemented system before execution.

ID	Scenario	Procedure / expected result	Status
ST-01	Role login and dashboard	Log in as student, warden, and administrator. Each role reaches the appropriate dashboard and sees role-appropriate navigation/data.	Not run
ST-02	Invalid login and logout	Attempt invalid credentials, then log in and log out. Invalid login is rejected with a clear message; logout ends authenticated access.	Not run
ST-03	Student application submission	Complete a valid application with required test data. Confirmation is shown and exactly one application is stored with the expected initial status.	Not run
ST-04	Application validation	Submit missing, malformed, and boundary values. Invalid data is identified and not saved; valid values are accepted.	Not run
ST-05	Application review decision	As authorized staff, review and approve/reject a pending application. The status, decision details, and student-visible result update correctly.	Not run
ST-06	Application search/filter	Search/filter applications by supported fields and statuses. Returned records match the criteria and counts/details remain consistent.	Not run
ST-07	Bed availability search	Search available rooms/beds using supported hostel and room criteria. Results reflect current occupancy and availability rules.	Not run
ST-08	Bed allotment	Allot an available bed to an eligible student. Allocation is saved, occupancy updates, and conflicting active allotments are prevented.	Not run
ST-09	Allotment change/cancellation	Change or cancel an allotment through the supported process. Student and bed status update consistently without orphaned or duplicate allocation.	Not run
ST-10	Complaint lifecycle	Submit a complaint, view it as the student, and update/close it as authorized staff. Status and history remain consistent across views.	Not run
ST-11	Fee status workflow	View a test student fee record and perform supported verification/update. Correct status and history appear to authorized users.	Not run
ST-12	Visitor entry and exit	Record a visitor entry then exit using valid test details. The record is searchable and the visit state/time data is accurate.	Not run
ST-13	Notice publishing and visibility	Create/publish a test notice if supported. Intended student audience can view it; unpublished or restricted notices remain hidden.	Not run
ST-14	Occupancy report reconciliation	Generate the occupancy report for test hostels. Totals reconcile with active allotments and available beds under the agreed counting rules.	Not run
ST-15	Dashboard/report consistency	After representative record changes, compare dashboards, details, search results, and reports. All views reflect the same current data.	Not run
ST-16	Duplicate submission handling	Repeat a submission or action through refresh/retry where applicable. The system avoids unintended duplicate applications, payments, or allotments.	Not run
ST-17	Concurrent bed allocation	Attempt allocation of the same bed to two eligible test students in a controlled run. At most one active allocation succeeds and the other receives a clear outcome.	Not run
ST-18	Error and recovery behavior	Use unavailable/invalid test records and recover from interrupted form submission. The system shows actionable errors and does not leave inconsistent partial records.	Not run

6. Acceptance Criteria

• All agreed critical end-to-end workflows complete with expected results.

• No unresolved critical workflow defects remain; severity and release thresholds are agreed with the project owner.

• Role-specific actions and record visibility match documented requirements.

• Application statuses, allotments, complaint states, fee states, visitor records, and reports remain consistent across relevant screens.

• No duplicate active bed allocation or unintended duplicate transaction is introduced by retry or concurrent action.

• Every defect has reproducible steps, evidence, severity, owner, and retest outcome.

7. Results and Measurements

No test execution output or measured system behavior was supplied. This section is intentionally pending and must be completed after running the cases in the agreed environment. Do not assign an overall pass based only on the presence of screens.

Run / case group	Passed	Failed	Blocked / not run	Evidence / notes
Core workflows / ST-01 to ST-06	___	___	___	_____
Allocation and services / ST-07 to ST-13	___	___	___	_____
Reports and integration / ST-14 to ST-18	___	___	___	_____

7.1 Defect and Observation Log

ID	Finding / evidence	Severity	Owner	Status
SYS-01	To be completed after execution	TBD	TBD	Open
SYS-02	Add a row for each distinct defect	TBD	TBD	Open

8. Execution Notes and Report Limits

This report is a proposed system test plan based on the project workflows named in the supplied performance-testing report. It does not establish functional requirements, implementation behavior, or actual test outcomes. Review the cases with the project supervisor, update expected behavior where needed, and record execution evidence before making release decisions.

9. Sign-off

Role	Name	Signature / date
Tester	_____	_____
Project supervisor	_____	_____
System owner / approver	_____	_____